SOLUTION OF HOMEWORK 06

1.

IDENTIFY: The mass and frequency are related by $f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$.

SET UP: $f\sqrt{m} = \frac{\sqrt{k}}{2\pi} =$ constant, so $f_1\sqrt{m_1} = f_2\sqrt{m_2}$.

EXECUTE: **(a)** $m_1 = 0.750$ kg, $f_1 = 1.33$ Hz and $m_2 = 0.750\text{ kg} + 0.220\text{ kg} = 0.970$ kg.

$$f_2 = f_1\sqrt{\frac{m_1}{m_2}} = (1.33\text{ Hz})\sqrt{\frac{0.750\text{ kg}}{0.970\text{ kg}}} = 1.17\text{ Hz}.$$

(b) $m_2 = 0.750\text{ kg} - 0.220\text{ kg} = 0.530$ kg. $f_2 = (1.33\text{ Hz})\sqrt{\frac{0.750\text{ kg}}{0.530\text{ kg}}} = 1.58$ Hz

EVALUATE: When the mass increases the frequency decreases and when the mass decreases the frequency increases.

2.

IDENTIFY: The motion is SHM, and in each case the motion described is one-half of a complete cycle.

SET UP: For SHM, $x = A\cos(\omega t)$ and $\omega = \frac{2\pi}{T}$.

EXECUTE: **(a)** The time is half a period. The period is independent of the amplitude, so it still takes 2.70 s.

(b) $x = 0.090$ m at time t_1. $T = 5.40$ s and $\omega = \frac{2\pi}{T} = 1.164$ rad/s. $x_1 = A\cos(\omega t_1)$. $\cos(\omega t_1) = 0.500$.

$\omega t_1 = 1.047$ rad and $t_1 = 0.8997$ s. $x = -0.090$ m at time t_2. $\cos(\omega t_2) = -0.500$ m. $\omega t_2 = 2.094$ rad and $t_2 = 1.800$ s. The elapsed time is $t_2 - t_1 = 1.800\text{ s} - 0.8997\text{ s} = 0.900$ s.

EVALUATE: It takes less time to travel from ± 0.090 m in (b) than it originally did because the block has larger speed at ± 0.090 m with the increased amplitude.

3.

IDENTIFY: Velocity and position are related by $E = \frac{1}{2}kA^2 = \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2$. Acceleration and position are related by $-kx = ma_x$.

SET UP: The maximum speed is at $x = 0$ and the maximum magnitude of acceleration is at $x = \pm A$.

EXECUTE: **(a)** For $x = 0$, $\frac{1}{2}mv_{\max}^2 = \frac{1}{2}kA^2$ and $v_{\max} = A\sqrt{\frac{k}{m}} = (0.040\text{ m})\sqrt{\frac{450\text{ N/m}}{0.500\text{ kg}}} = 1.20$ m/s

(b) $v_x = \pm\sqrt{\frac{k}{m}}\sqrt{A^2 - x^2} = \pm\sqrt{\frac{450\text{ N/m}}{0.500\text{ kg}}}\sqrt{(0.040\text{ m})^2 - (0.015\text{ m})^2} = \pm 1.11$ m/s.

The speed is $v = 1.11$ m/s.

(c) For $x = \pm A$, $a_{\max} = \frac{k}{m}A = \left(\frac{450\text{ N/m}}{0.500\text{ kg}}\right)(0.040\text{ m}) = 36\text{ m/s}^2$

(d) $a_x = -\frac{kx}{m} = -\frac{(450\text{ N/m})(-0.015\text{ m})}{0.500\text{ kg}} = +13.5\text{ m/s}^2$

(e) $E = \frac{1}{2}kA^2 = \frac{1}{2}(450\text{ N/m})(0.040\text{ m})^2 = 0.360$ J

EVALUATE: The speed and acceleration at $x = -0.015$ m are less than their maximum values.

4.

IDENTIFY: The graph shows that the amplitude of vibration is decreasing, so the system must be losing mechanical energy.

SET UP: The mechanical energy is $E = \frac{1}{2}mv_x^2 + \frac{1}{2}kx^2$.

EXECUTE: **(a)** When $|x|$ is a maximum and the tangent to the curve is horizontal the speed of the mass is zero. This occurs at $t = 0$, $t = 1.0$ s, $t = 2.0$ s, $t = 3.0$ s and $t = 4.0$ s.

(b) At $t = 0$, $v_x = 0$ and $x = 7.0$ cm so $E_0 = \frac{1}{2}kx^2 = \frac{1}{2}(225\text{ N/m})(0.070\text{ m})^2 = 0.55$ J.

(c) At $t = 1.0$ s, $v_x = 0$ and $x = -6.0$ cm so $E_1 = \frac{1}{2}kx^2 = \frac{1}{2}(225\text{ N/m})(-0.060\text{ m})^2 = 0.405$ J.

At $t = 4.0$ s, $v_x = 0$ and $x = 3.0$ cm so $E_4 = \frac{1}{2}kx^2 = \frac{1}{2}(225\text{ N/m})(0.030\text{ m})^2 = 0.101$ J. The mechanical energy "lost" is $E_1 - E_4 = 0.30$ J. The mechanical energy lost was converted to other forms of energy by nonconservative forces, such as friction, air resistance and other dissipative forces.

EVALUATE: After a while the mass will come to rest and then all its initial mechanical energy will have been "lost" because it will have been converted to other forms of energy by nonconservative forces.

5.

After landing on a strange new planet, a space explorer constructs a simple pendulum of length 55.0cm. The astronaut finds that the pendulum makes 105 complete swings in 135s. What is the value of g on this planet?

- 13.1m/s^2
- 13.1cm/s^2
- 7.5cm/s^2
- 7.5m/s^2

6.

At a distance of 9.2×10^{12}m from a star, the stars power output is estimated to be 1.6×10^{28} W. If we assume the star radiates uniformly in all directions, what is the intensity of the star?

Give your answer in units of W/m^2, to 1d.p.

15 margin of error +/- 2%

7.

Waves transport energy, but not matter, from one region to another.

Is this true or false?

- True
- False

8.

Example 15.1 Wavelength of a musical sound

Sound waves are longitudinal waves in air. The speed of sound depends on temperature; at 20°C it is 344 m/s (1130 ft/s). What is the wavelength of a sound wave in air at 20°C if the frequency is 262 Hz (the approximate frequency of middle C on a piano)?

SOLUTION

IDENTIFY and SET UP: This problem involves Eq. (15.1), $v = \lambda f$, which relates wave speed v, wavelength λ, and frequency f for a periodic wave. The target variable is the wavelength λ. We are given $v = 344$ m/s and $f = 262$ Hz $= 262\ \text{s}^{-1}$.

EXECUTE: We solve Eq. (15.1) for λ:

$$\lambda = \frac{v}{f} = \frac{344\ \text{m/s}}{262\ \text{Hz}} = \frac{344\ \text{m/s}}{262\ \text{s}^{-1}} = 1.31\ \text{m}$$

EVALUATE: The speed v of sound waves does *not* depend on the frequency. Hence $\lambda = v/f$ says that wavelength changes in inverse proportion to frequency. As an example, high (soprano) C is two octaves above middle C. Each octave corresponds to a factor of 2 in frequency, so the frequency of high C is four times that of middle C: $f = 4(262\ \text{Hz}) = 1048$ Hz. Hence the *wavelength* of high C is *one-fourth* as large: $\lambda = (1.31\ \text{m})/4 = 0.328$ m.

9.

Example 15.2 Wave on a clothesline

Cousin Throckmorton holds one end of the clothesline taut and wiggles it up and down sinusoidally with frequency 2.00 Hz and amplitude 0.075 m. The wave speed on the clothesline is $v = 12.0$ m/s. At $t = 0$ Throcky's end has maximum positive displacement and is instantaneously at rest. Assume that no wave bounces back from the far end. (a) Find the wave amplitude A, angular frequency ω, period T, wavelength λ, and wave number k. (b) Write a wave function describing the wave. (c) Write equations for the displacement, as a function of time, of Throcky's end of the clothesline and of a point 3.00 m from that end.

SOLUTION

IDENTIFY and SET UP: This is a kinematics problem about the clothesline's wave motion. Throcky produces a sinusoidal wave that propagates along the clothesline, so we can use all of the expressions of this section. In part (a) our target variables are A, ω, T, λ, and k. We use the relationships $\omega = 2\pi f$, $f = 1/T$, $v = \lambda f$, and $k = 2\pi/\lambda$. In parts (b) and (c) our target "variables" are expressions for displacement, which we'll obtain from an appropriate equation for the wave function. We take the positive x-direction to be the direction in which the wave propagates, so either Eq. (15.4) or (15.7) will yield the desired expression. A photograph of the clothesline at time $t = 0$ would look like Fig. 15.9a, with the maximum displacement at $x = 0$ (the end that Throcky holds).

EXECUTE: (a) The wave amplitude and frequency are the same as for the oscillations of Throcky's end of the clothesline, $A = 0.075$ m and $f = 2.00$ Hz. Hence

$$\omega = 2\pi f = \left(2\pi \frac{\text{rad}}{\text{cycle}}\right)\left(2.00 \frac{\text{cycles}}{\text{s}}\right)$$
$$= 4.00\pi \text{ rad/s} = 12.6 \text{ rad/s}$$

The period is $T = 1/f = 0.500$ s, and from Eq. (15.1),

$$\lambda = \frac{v}{f} = \frac{12.0 \text{ m/s}}{2.00 \text{ s}^{-1}} = 6.00 \text{ m}$$

We find the wave number from Eq. (15.5) or (15.6):

$$k = \frac{2\pi}{\lambda} = \frac{2\pi \text{ rad}}{6.00 \text{ m}} = 1.05 \text{ rad/m}$$

or

$$k = \frac{\omega}{v} = \frac{4.00\pi \text{ rad/s}}{12.0 \text{ m/s}} = 1.05 \text{ rad/m}$$

(b) We write the wave function using Eq. (15.4) and the values of A, T, and λ from part (a):

$$y(x,t) = A\cos 2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)$$
$$= (0.075 \text{ m})\cos 2\pi\left(\frac{x}{6.00 \text{ m}} - \frac{t}{0.500 \text{ s}}\right)$$
$$= (0.075 \text{ m})\cos[(1.05 \text{ rad/m})x - (12.6 \text{ rad/s})t]$$

We can also get this same expression from Eq. (15.7) by using the values of ω and k from part (a).

(c) We can find the displacement as a function of time at $x = 0$ and $x = +3.00$ m by substituting these values into the wave function from part (b):

$$y(x = 0, t) = (0.075 \text{ m})\cos 2\pi\left(\frac{0}{6.00 \text{ m}} - \frac{t}{0.500 \text{ s}}\right)$$
$$= (0.075 \text{ m})\cos(12.6 \text{ rad/s})t$$
$$y(x = +3.00 \text{ m}, t) = (0.075 \text{ m})\cos 2\pi\left(\frac{3.00 \text{ m}}{6.00 \text{ m}} - \frac{t}{0.500 \text{ s}}\right)$$
$$= (0.075 \text{ m})\cos[\pi - (12.6 \text{ rad/s})t]$$
$$= -(0.075 \text{ m})\cos(12.6 \text{ rad/s})t$$

EVALUATE: In part (b), the quantity $(1.05 \text{ rad/m})x - (12.6 \text{ rad/s})t$ is the *phase* of a point x on the string at time t. The two points in part (c) oscillate in SHM with the same frequency and amplitude, but their oscillations differ in phase by $(1.05 \text{ rad/m})(3.00 \text{ m}) = 3.15$ rad $= \pi$ radians—that is, one-half cycle—because the points are separated by one half-wavelength: $\lambda/2 = (6.00 \text{ m})/2 = 3.00$ m. Thus, while a graph of y versus t for the point at $x = 0$ is a cosine curve (like Fig. 15.9b), a graph of y versus t for the point $x = 3.00$ m is a *negative* cosine curve (the same as a cosine curve shifted by one half-cycle).

Using the expression for $y(x = 0, t)$ in part (c), can you show that the end of the string at $x = 0$ is instantaneously at rest at $t = 0$, as stated at the beginning of this example? (*Hint:* Calculate the y-velocity at this point by taking the derivative of y with respect to t.)

10.

find.

$A = 0.01\,m$, $\omega = \frac{2\pi}{T} = 200\pi$; $\lambda = u \cdot T = 4\,m$; $\varphi = -\frac{\pi}{2}$

wave source.

Solution: (a) $y(x,t) = 0.01\cos(200\pi t - \frac{\pi}{2}x - \frac{\pi}{2})$